From Relation to Node:

Minimal Conditions for the Emergence of Grounded Symbolic Graphs, with a Machine-Verified Formalization in Lean 4

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Status. This document is a work in progress. It is incomplete, and all of its parts, including the axioms, definitions, results and their interpretation, are subject to revision and refinement. Passages marked in the margin were drafted with AI assistance and have not yet been checked against their sources.

Mysteries should not be multiplied beyond necessity.
— Eric Gans, *The Origin of Language* (1981)

1 Motivation and Background

1.1 Theories of language origin

Theories of language origin attempt to explain how a species without language came to have it. The field is broad. Among its main families are the following.

- Gestural accounts, on which language grew out of manual communication and pantomime [4, 2].
- Social accounts, on which language took over the role of grooming in maintaining group bonds [8].
- Intention-based accounts, which ground meaning in the expression and recognition of communicative intentions [13, 25].
- Game-theoretic accounts, in which signals acquire meaning as stable conventions among interacting agents [17, 23].
- Computational accounts, which locate the decisive step in the appearance of a single combinatorial operation [3].

Two problems recur across these families, and they help explain why the question of language origin has often been treated with suspicion.

Presupposition. Many accounts take as given, in some form, capacities closely related to the ones whose origin is at issue. This is not always an oversight; often it is a deliberate choice of what is to be explained. Three examples illustrate the point.

- *Game-theoretic accounts.* Lewis–Skyrms signaling games show how signals can acquire meaning without design or prior agreement. Skyrms extends the

framework so that agents begin with no signals at all and invent new ones through a reinforcement process [23, 1]. What these models take as given is the interaction itself: a sender whose behaviour depends on a state of the world, a receiver whose action depends on the sender's behaviour, and payoffs that reward a match. The models explain which behaviour comes to mean which state. That one participant's behaviour is the kind of thing the other's action responds to *as* a signal is the frame within which they work.

- *Intention-based accounts.* These analyze meaning through the expression and recognition of communicative intentions [13]. As analyses of meaning, they are not committed to any story of origin. As a basis for one, they inherit the observation that an intention to communicate is itself a representational state, so that thought is presupposed prior to language. Tomasello's account is explicit on this point. It posits shared intentionality, joint attention and cooperative motives as a biological infrastructure that evolved before conventional language and made it possible [25]. The question of reference is moved to the evolution of that infrastructure, not answered from nothing.

Unverified (AI-drafted). The thought-priority objection to intention-based accounts needs a citation. A candidate is Devitt, "Three Mistakes About Semantic Intentions".

- *Computational accounts.* Berwick and Chomsky locate the decisive innovation in a single combinatorial operation, Merge, and explain how hierarchical structure is built from lexical atoms [3]. They accept that words preceded grammar, and critics have pointed out that the origin of those atoms is left largely unaddressed [24].

None of these observations is a refutation. Each account addresses its own question, and every model must take something as given, including the one developed here, which assumes perception and response. The point concerns where the line is drawn. In each case, the step by which some piece of behaviour first comes to *stand for* something is not part of what is explained.

Evidence. Language leaves no fossils. The events in question are prehistoric, unobservable and unrepeatable, and the indirect evidence from anatomy, archaeology, comparative cognition and genetics constrains them only weakly. In 1866 the Société de Linguistique de Paris went as far as to exclude communications on the origin of language from its proceedings [18]. Much more recently, a review by leading researchers concluded that the central questions remain largely out of reach of the available evidence [14]. Theories of origin are therefore often regarded as untestable, and dismissed as just-so stories: narratives that fit what they explain, but that no observation could refute.

These two problems point toward a particular kind of contribution. A theory of origin that cannot be tested historically may still be assessed in another way: by asking whether its conclusions *follow* from its assumptions, and which assumptions they require. A claim of the form "if these conditions hold, this structure can arise", or "must have this form", is not a claim about what happened. It can be checked deductively, and its assumptions can then be examined individually, some of them empirically, for plausibility. This does not make an origin story testable as history. It makes it accountable as an argument.

The originary hypothesis. Eric Gans's Generative Anthropology is notable for addressing the first problem directly, and in a way that lends itself to this kind of assessment [9, 10]. Several features distinguish it.

1. It starts from the sign itself. It asks what must happen for a sign to exist at all, not how pre-existing contents come to be communicated.

2. It derives the sign from non-semiotic behaviour. On Gans's account, the first sign is an aborted gesture of appropriation: a practical act that, by being interrupted, comes to designate instead of to seize. This gives a concrete mechanism by which representation could arise without being presupposed.

3. The sign is shared from the outset. It arises in a scene among several participants, as a mutually understood gesture, so meaning is not a private state later made public.

4. It is generative. A single minimal hypothesis is followed by a derivation of further elementary forms of language in sequence. A claimed derivation has internal structure that can be examined step by step.

The Origin of Language opens with the sentence "Mysteries should not be multiplied beyond necessity" [9]. It is a version of Ockham's razor, and it states the ambition of the theory: to explain the origin of the human while assuming as little as possible.

The hypothesis runs as follows. A group of hominids converge, through mimetic desire, on a single desirable object. Mutual imitation intensifies their rivalry beyond what an animal dominance hierarchy can contain, and the group faces a violent crisis. The crisis is deferred when the gesture of appropriation is aborted. The aborted gesture, shared and reciprocally understood, becomes the first sign: an *ostensive* that designates the object as a centre which no one may take. The scene concludes in the *sparagmos*, the collective division of the object.

Unverified (AI-drafted). The role of the dominance hierarchy in Gans's scene has not been confirmed in a source. Check Gans 1981 or 1993.

From the ostensive, Gans derives the further forms in sequence. The *imperative* arises from an ostensive made in the absence of its referent, which the hearer answers by supplying the missing object. The *declarative* arises when the imperative cannot be fulfilled, and the hearer responds with a negation of the object's presence.

Unverified (AI-drafted). This summary of the imperative (the "inappropriate ostensive") and the declarative (the negative ostensive) was confirmed only through secondary commentary. Check against Gans 1981.

Later work in the tradition develops related notions. Adam Katz, who also writes as Dennis Bouvard, introduced into the originary scene the *firstness* of a single participant who emits the sign before the others [16]; see also [11]. He has also developed an approach organized around the *centre* toward which the attention of a scene converges.

Unverified (AI-drafted). Katz on the centre has not been read directly. A candidate source is Katz, "Generative Anthropology as the One Big Discipline", *Anthropoetics* XXIV, no. 2 (2019).

1.2 The problem: narrative content of uncertain necessity

The present work grows out of a long engagement with the writings of Gans and Katz, and out of a specific dissatisfaction with one feature of the account. Gans's hypothesis avoids presupposing the sign. The scene that does this work, however, carries substantial narrative content:

- *convergence*: mimetic desire draws the participants to a single object;
- *crisis*: mimetic rivalry exceeds what an animal dominance hierarchy can contain;

- *deferral*: the prospect of violence motivates the abortion of the gesture of appropriation;
- *centre*: the designated object becomes a sacred centre;
- *resolution*: the scene ends in the sparagmos, in which the tension the sign has deferred is discharged on the object, not on other members of the group.

Much of this content is inherited from René Girard's mimetic theory [12]. Gans accepts that language originates in a mimetic crisis, while rejecting the scapegoat mechanism as sufficient to produce a sign. The sparagmos is in part his way of incorporating Girard's account of sacrificial violence. Each element is anthropologically suggestive, and each ties the origin of language to themes the theory develops elsewhere, such as the sacred, ritual, ethics and exchange.

Which of these elements is actually needed is a natural question for an account that begins by refusing to multiply mysteries. It has been raised within the tradition itself. Sackeroff, for example, identifies the mechanism of deferral as a point of difficulty, observing that the account draws on two different sources of motivation: animal appetite for the object, and a distinctively human investment in the sign [22]. The present work takes up this internal question in a formal setting.

Several of the scene's elements are open to objection on their own terms.

- *Violence.* Among primates, conflict is frequent but usually limited, and it is often followed by reconciliation [6]. Evolutionary game theory explains why contests between members of the same species usually take the form of limited, not escalated, fighting [19]. Disputes that end in exhaustion, hierarchy or reconciliation do not by themselves produce signs. The account's answer is that mimetic intensity has disabled these ordinary resolutions, which is itself one of the elements whose necessity is in question.

- *Deferral as a strategy.* In the Hawk–Dove game, mutual restraint is not stable: against a participant who refrains, taking the resource does better. A deferral that rests on self-interest therefore needs further explanation. The account supplies it through the intensity of mimetic desire and the threat of collective violence, again through the narrative content.

- *Division.* By the time the sparagmos begins, the aborted gesture has already designated the object. Its role in the origin of distribution and of the sacred is not in question here. Its role in the emergence of reference is unclear.

The issue is therefore not whether these elements are plausible. It is whether they are *load-bearing*. If the derivation of the sign and its forms depends on them, it inherits their contingency, and the charge of a just-so story applies. If it does not, the derivation should be statable without them.

The present work follows in the spirit of Gans's opening sentence, and applies it to the originary scene itself. Mimetic rivalry, catastrophic violence and the sparagmos are explanatory commitments, and each is a candidate for a mystery multiplied beyond necessity. The question is whether the derivation of the sign and its forms requires them. If it does not, the principle with which Gans begins recommends setting them aside.

Unverified (AI-drafted). This description of the sparagmos comes from an *Anthropoetics* page whose author has not been confirmed. Find a citable source.

Unverified (AI-drafted). The relation between Gans and Girard summarized here has been confirmed only through an encyclopedia entry. Find a primary or scholarly source.

Unverified (AI-drafted). The issue number of Sackeroff's article was inferred from its URL. Confirm.

Unverified (AI-drafted). The Hawk–Dove game postdates Maynard Smith and Price (1973). Cite the right source, probably Maynard Smith, *Evolution and the Theory of Games* (1982).

1.3 The target: a grounded symbolic graph

A generative account needs a destination as well as a starting point. Both can be stated using accepted notions of the sign.

The starting state. There are actors and physical objects. There are contexts: at each time, which actors are present and which objects are within whose reach. Actors move, and some movements are directed at objects. All the relations in this state are momentary and local. Each holds between particular actors and things, at a particular time, in a particular context. There are no symbolic nodes.

The sign as a relation. The accepted definitions of the sign are relational. For Saussure, the linguistic sign joins a concept and an acoustic image, and its value is fixed only by its differences from other signs in the system. For Peirce, "a sign, or representamen, is something which stands to somebody for something in some respect or capacity" [21, 2.228]. This is a relation among a sign, an object and an interpretant, and it holds *for* someone. Morris divides the study of signs by the relations a sign can enter into: to other signs, to objects, and to its interpreters [20].

Unverified (AI-drafted). Add a citation, with page, to the *Course in General Linguistics*.

On each account, a sign is not a thing with a property but a term in relations. Among those relations are relations to the *actors* who produce and interpret it. The minimal mathematical structure of relations is a graph. Relations among more than two terms can be represented as a hypergraph, or encoded in an ordinary graph by an incidence construction, in which each instance of a relation becomes a node joined to its arguments. The relevant graphs are many-sorted: their vertices include actors, physical objects, and the acts by which actors relate to objects and to one another.

Actors, time and context. Every such relation is indexed by actors, by a time and by a context. The most elementary sign relation, ostension, shows all three at once. *This*, *here* and *now* designate only within the situation in which they are produced, for those who share it. A pointing gesture relates one actor, by way of a movement, to an object, at one moment and in one place, and only for another actor who takes it up. The relation does not outlast the act.

Symbolic reference is what escapes this indexing. Hockett lists *displacement*, the capacity to refer to what is removed in time or place from the moment of speaking, among the design features of human language [15]. Deacon, building on Peirce, argues that symbolic reference is grounded in indexical reference but depends on relations among symbols themselves [7]. In graph terms, a symbolic node is something that persists across contexts and across occasions of use: different actors can re-identify it, at other times and in other places, and it can enter into further relations there.

The end state. A generative account must arrive at a graph with the following properties.

1. *Nodes that stand for things.* Some vertices are symbolic, and stand for objects to actors.

2. *Persistence beyond the context of origin.* A symbolic node can be re-identified at another time, in another place and by another actor, including in the absence of what it stands for.

3. *Relations among symbolic nodes.* Symbolic nodes stand in relations to one another, not only to objects, and some of these relations have *roles*, where it matters which term is which.

4. *Grounding.* Every symbolic node can be traced back to the physical world, so there are no free-floating symbols.

5. *Truth and falsity.* What is said with symbolic nodes has conditions under which it holds, and it can fail.

6. *Partiality.* Relations can be affirmed, denied, or left unsaid, without silence having to carry information.

7. *Open-endedness.* Any finite relational structure over grounded objects can be expressed.

Unverified (AI-drafted). Hockett and Altmann's "prevarication" may be the right citation here. Check.

The generative task. The task is to get from the starting state to the end state using only what the starting state contains, and operations on it applied to their own products. Nothing symbolic may be assumed along the way. Several features commonly attributed to human language lie outside the target: the vocal-auditory channel, duality of patterning, and cultural transmission across generations. The account addresses the emergence of reference and of relational structure, not of speech.

The target also makes precise a hypothesis that motivated the work:

- the ostensive establishes an edge between actors and an object within a single context;
- the imperative is such an edge whose object is missing from the actors' current context;
- the declarative yields a node: a relation that has itself become something the actors can refer to, and that persists across contexts.

The formalization supports a sharper version of the last point. The first symbolic node is a *reified relation*, and relations between nodes arise by the same operation applied one level up.

1.4 Aim: a minimal axiom set that does not license itself

The aim is to find a minimal set of axioms under which the end state of Section 1.3 can be reached from the starting state, and to prove that it can.

The rule against self-licensing. An axiom set *licenses itself* when what it is supposed to generate is built into its assumptions, so that the derivation only returns what was put in. To exclude this, every axiom set considered here must satisfy two conditions.

1. *Base vocabulary only.* Axioms may speak only of what the starting state contains: actors, physical objects, times, movements, reach, contact, perception,

and resemblance between movements. They may not mention signs, meaning, reference, uptake, the ostensive, the imperative, the declarative, symbolic nodes, claims or truth, nor any notion defined from these.

2. *Target notions by definition only.* Every notion that belongs to the end state is introduced by an explicit definition in the base vocabulary. A definition adds no constraint on what can happen: it names a pattern, which may or may not occur.

Under these conditions, every property of the end state that the account establishes is a theorem, derived from axioms that do not mention it.

Definitions are hypotheses. The rule shifts the weight of the account onto its definitions: what it is for a movement to be taken up as being about something, or for a pattern of responses to constitute a shared centre. A definition can smuggle in as much as an axiom can. Two safeguards apply. Definitions are drawn as closely as possible from accepted notions of the sign, and they are stated as explicit hypotheses about what the defined notion consists in, open to dispute and to replacement by alternatives.

Generativity relative to the target. With the target fixed and the rule in force, the aim has three parts.

Existence.
There is a model of the axioms in which the end state, or the relevant part of it, arises from a starting state that lacks it.

Necessity.
In every model of the axioms, whenever part of the end state exists, it has a determinate origin and order.

Universality.
The operations that produce the first symbolic structures suffice for open-endedness: any finite relational structure over grounded objects can be reached.

Minimality. An axiom set is minimal when each axiom is needed: for each axiom there is a model of the others in which some property of the end state fails. Minimality is shown axiom by axiom, by exhibiting such countermodels.

The aim is not to show that the end state was inevitable, nor that it was reached in one particular way. It is to show that it can be reached under assumptions that are weak, explicit and free of the target, and to identify which assumptions each part of it requires.

Comparing accounts. The same rule suggests a way of comparing accounts. Formalize an account, separate its primitives into base vocabulary and anything semiotic, and ask whether its target notions can be defined from the base part alone. Padoa's method from logic gives a standard test: a notion is not definable from the other primitives if two models agree on all of them but disagree on that notion. One feature that may signal self-licensing is a primitive kind of act whose

only role is to be responded to, such as the costless signals of a signaling game. Any such result holds only relative to a particular formalization, and only relative to the present target, the emergence of sign-hood. It is not a criticism of what other accounts set out to explain. Applying the test to other accounts, including Gans's own scene, is left as work in progress.

1.5 Rival orderings

A further motivation is comparative. The sequence ostensive, imperative, declarative is one reading among several, and other readings, for instance ones in which the declarative is primary, are possible. Under the rule of Section 1.4, the choice between them becomes precise. An account that builds a particular mode into its axioms licenses its own ordering. The real question is which orderings can be generated from axioms in the base vocabulary, and under which assumptions. The framework developed here bears on this directly. In it, the declarative requires the *absence* of its object, but not a prior failed imperative. The imperative and the declarative appear as two responses to absence, not as successive stages.

Unverified (AI-drafted). Name proponents of declarative-first or other orderings, or rephrase as the author's conjecture.

1.6 Why machine verification

Arguments in humanistic theory often rely on steps that are intuitively compelling but unstated. Formalization in a proof assistant forces every assumption to be written down and every inference to be checked. All results reported here are formalized in the Lean 4 proof assistant [5] and checked in full, with no unproven steps.

The process was itself informative. Several assumptions that the informal narrative leaves implicit were exposed only when a proof failed without them. Among them:

- a kinematic principle linking a reproduced gesture to the object it was originally directed at;
- the requirement that one participant act before another, which corresponds to Katz's firstness;
- the dependence of falsehood on the drift of signs away from their iconic origins;
- the reliance of affirmation, derived as the negation of a negation, on classical logic.

Formalization also makes it possible to say exactly which statements are derived and which are merely definitions. The paper marks this distinction throughout.

1.7 Outline

Section 2 introduces the primitives and axioms, which concern only actors, movements, reach and resemblance. Section 3 defines taking-as, the centre, dependent response and the elementary modes as patterns of response. Section 4 reports the results that hold in every model: the origin of signs, the form of the imperative, the declarative's dependence on absence, direction of fit, and the conditions for falsehood. Section 5 reports the results on graphs: claims, universality, roles and polarity. Section 6 reports the models and countermodels that establish existence

and show which assumptions are needed. Section 7 discusses the relation to Generative Anthropology, open problems, and directions for collaborative work. Table 1 shows where each property of the end state is reached.

Property of the end state	Where it is reached
1. Nodes that stand for things	declarative and lineage (Section 3)
2. Persistence	re-presentation; displaced claims (Sections 4–5)
3. Relations, with roles	claims and ordered claims (Section 5)
4. Grounding	content traces to an object once within reach (Section 4)
5. Truth and falsity	falsehood requires drift (Section 4)
6. Partiality	polarity; partial universality (Section 5)
7. Open-endedness	universality (Section 5)

Table 1: Where each property of the end state is reached.

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